Fantasy Football Assistant - Project Documentation

Problem Statement

Fantasy Premier League (FPL) managers face a challenging decision-making problem: select 15 players from a pool of 500+ within a £100M budget, subject to position constraints (2 GKP, 5 DEF, 5 MID, 5 FWD), maximum 3 players per club, while maximizing expected points across upcoming gameweeks.

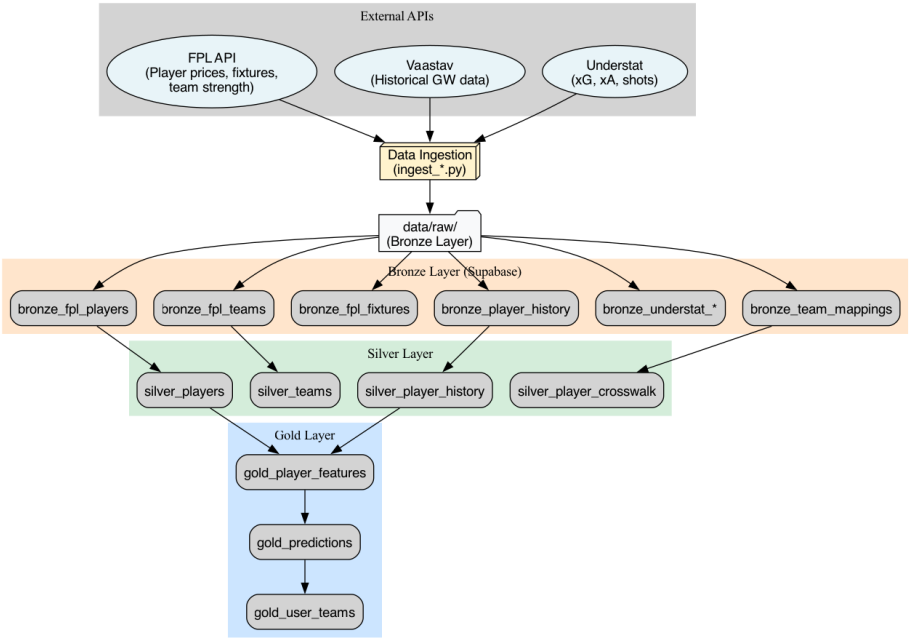
The core challenge: - Information overload: 500+ players, 38 gameweeks, multiple data sources - Complex constraints: budget, positions, club limits, captain choices - Uncertainty: player injuries, form fluctuations, fixture difficulty - Time-intensive: analyzing fixtures, form, matchups manually

Our approach: A data-driven system that automates prediction and optimization to help managers make better decisions.

The Role of AI in This Project

This project demonstrates how artificial intelligence can transform decision-making in competitive fantasy sports. With hundreds of players, constantly changing form, and complex constraints, manually analyzing all relevant factors is practically impossible for any human. AI solves this by processing historical data at scale, detecting patterns invisible to the human eye, and generating probabilistic forecasts that account for uncertainty. The LightGBM models learn from features like rolling performance, head-to-head records, fixture difficulty, and rest days to predict not just expected points, but also the probability of a player starting. Without AI, managers must rely on intuition or simple statistics; with AI, they gain data-driven confidence in their decisions. This architecture can be extended with future AI capabilities such as sentiment analysis on injury news and generative AI for natural language insights—further enhancing the assistant's value as an ongoing decision-support tool.

Data Sources



Data Sources Diagram

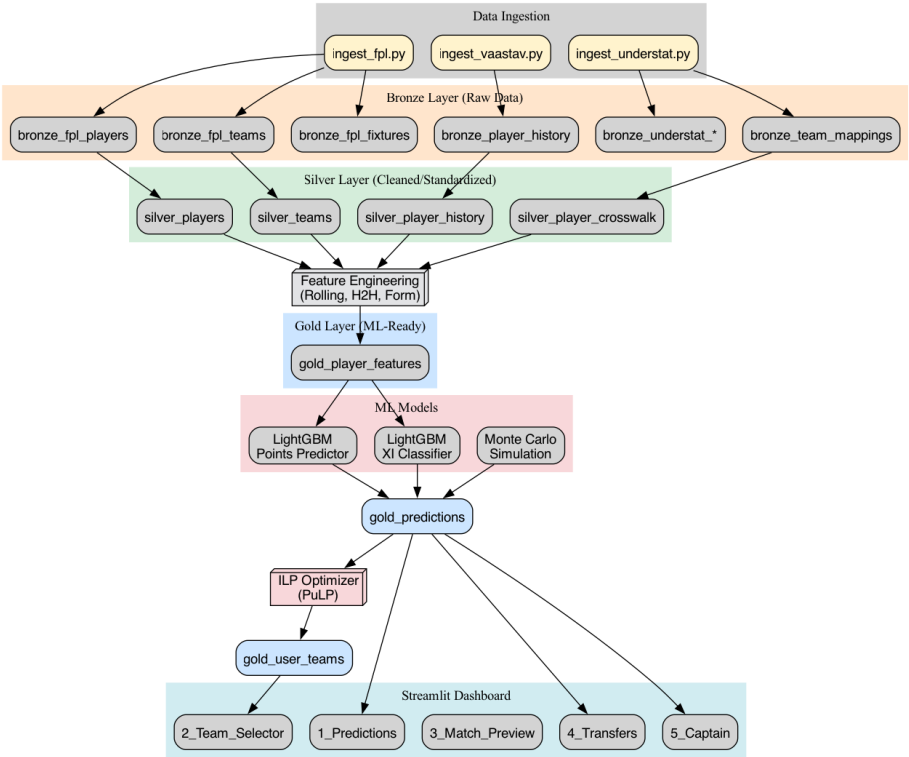
Source	Description	Data Type	Update Frequency
FPL API	Official Fantasy Premier League API	Player prices, selections, fixtures, team strength	Daily
Vaastav	Historical FPL gameweek data (2021-2025)	Per-player GW stats, points, minutes	One-time + annual
Understat	xG, xA, shot data	Match-level xG/xA, player match stats	Historical + seasonal
Team Mappings	Cross-source team ID mappings	FPL ? Understat ? Vaastav ID mappings	Auto-generated

Available Data Files

```
data/raw/
├── fpl/2025-26/
│   └── bootstrap-static.json    # Current season player/team data
├── vaastav/
│   ├── 2021-22/gws.parquet      # Historical GW data
│   ├── 2022-23/gws.parquet
│   ├── 2023-24/gws.parquet
│   └── 2024-25/gws.parquet
├── understat/
│   ├── 2021_22/                 # Player match stats, shots
```

```
│   ├── 2022_23/
│   ├── 2023_24/
│   ├── 2024_25/
│   └── 2025_26/
└── team_mappings.csv           # Cross-source mappings
```

Architecture & Medallion Structure

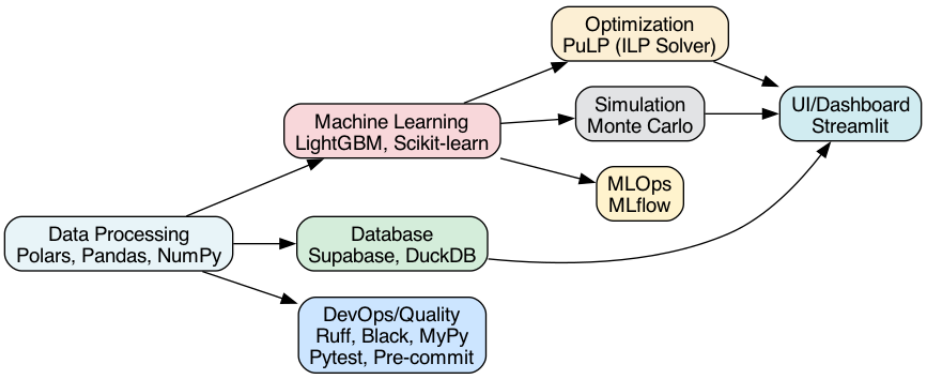


Architecture Diagram

Data Flow

Layer	Tables	Purpose
Bronze	`bronze_*`	Raw, immutable data as ingested
Silver	`silver_*`	Cleaned, standardized, crosswalk applied
Gold	`gold_*`	Features aggregated, predictions ready

Tools & Technologies

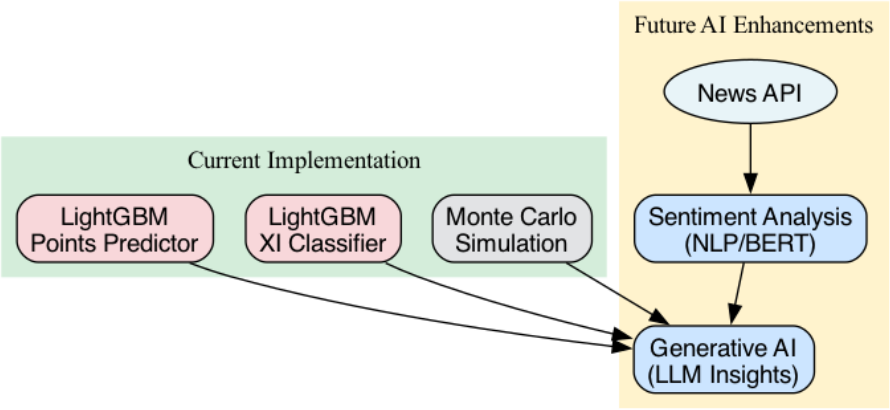


Tech Stack Diagram

High-Level Categories

Category	Tools	Purpose
Data Processing	Polars, Pandas, NumPy	ETL, feature engineering
ML Models	LightGBM, Scikit-learn, XGBoost	Performance prediction, classification
Optimization	PuLP (ILP solver)	Squad optimization under constraints
Simulation	NumPy (Monte Carlo)	Point distribution generation
Database	Supabase (Postgres), DuckDB	Storage, querying
MLOps	MLflow	Model tracking, versioning
Validation	Pandera, Pydantic	Schema enforcement
UI	Streamlit	Interactive dashboard
Code Quality	Ruff, Black, MyPy, Pre-commit	Linting, formatting, type checking
Testing	Pytest, Coverage	Unit/integration testing
CI/CD	GitHub Actions	Automated pipeline

AI & Machine Learning Implementation



AI Implementation Diagram

Current AI/ML Components

Component	Type	Description
Points Predictor	Regression (LightGBM)	Predicts expected points per gameweek
XI Classifier	Classification (LightGBM)	Predicts starting XI probability
Monte Carlo Simulator	Statistical Simulation	10,000 iterations for point distributions

Core ML Models

```
# Example: Points Prediction Model
from src.models.lightgbm_model import train_points_predictor

model = train_points_predictor(
    features_df,           # gold_player_features
    target="total_points",
    model_path="data/models/points_predictor.joblib"
)

# Example: XI Probability Model
from src.models.starting_xi import train_xi_model

xi_model = train_xi_model(
    features_df,
    target="minutes",
    threshold=60,
    model_path="data/models/xi_classifier.joblib"
)
```

Performance Targets

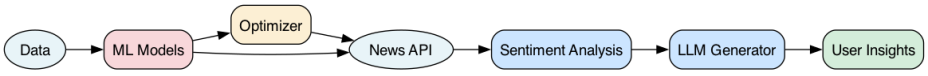
Model	Metric	Target
Points Predictor	MAE	< 2.0 points
Goals Predictor	RMSE	< 0.5 goals
XI Classifier	Accuracy	> 80%
Clean Sheet	Accuracy	> 75%

Future AI Implementations

Planned Enhancements (BACKLOG-003)

Feature	AI Type	Description
Sentiment Analysis	NLP	Analyze news articles, social media for player sentiment
Generative AI Insights	GenAI	Natural language transfer recommendations
LLM Match Previews	GenAI	Generate match narrative summaries
Injury Impact Analysis	NLP/Sentiment	Assess injury news impact on player value

Architecture for Future AI



Future AI Diagram

Suggested Stack for Future AI

Component	Option	Notes
Sentiment Analysis	HuggingFace Transformers	Pre-trained BERT for sentiment
LLM	OpenAI GPT-4 / Anthropic Claude / Ollama (local)	For insight generation
News API	NewsAPI, Twitter/X API	Player news ingestion

Streamlit Dashboard Preview

Current Pages

```
# src/streamlit_app/pages/
1_Predictions.py      # Player expected points & probabilities
2_Team_Selector.py   # Squad optimizer (ILP)
3_Match_Preview.py    # H2H analysis
4_Transfer_Suggestions.py  # Transfer recommendations
5_Captain_Analysis.py     # Captain picks
```

Page Wireframes

1. Predictions Page

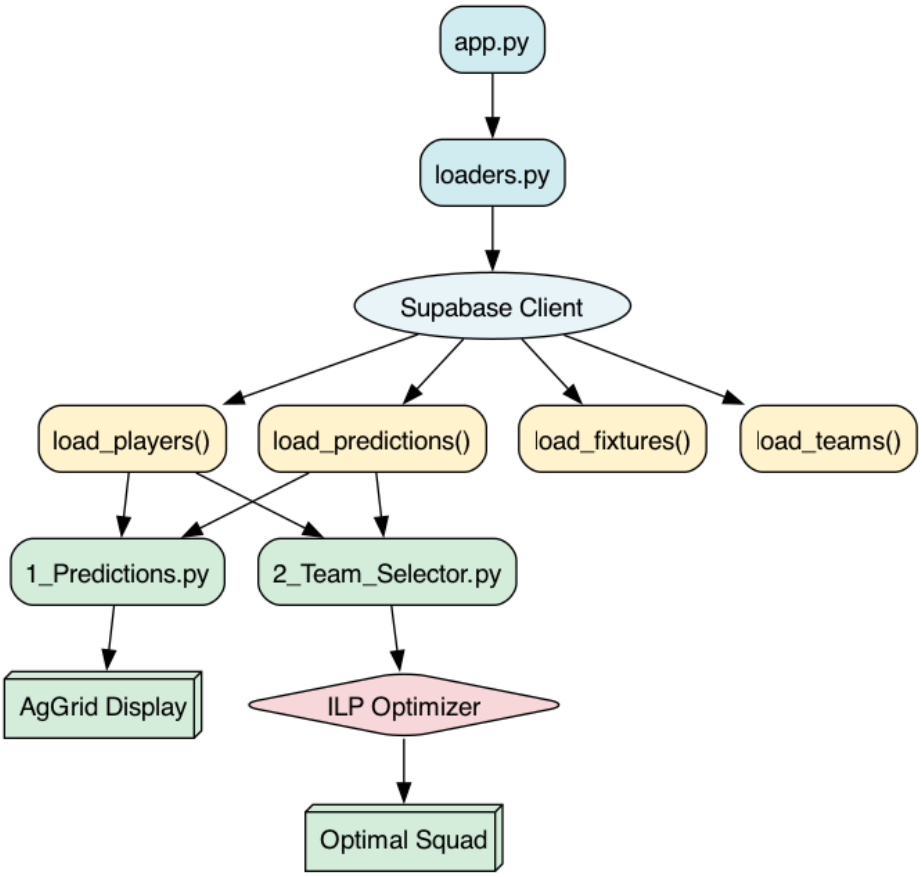
```
┌──────────────────────────────────────────────────────────────────────┐
│ ⚽ FPL Simulation Dashboard                          [Gameweek ▼]  │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│  ┌─────────────┐  ┌─────────────┐  ┌─────────────┐                  │
│  │ Filters     │  │ Metrics     │  │ Top Players │                  │
│  │             │  │             │  │             │                  │
│  │ Position ▼  │  │ Avg xPTS 4.2│  │ 1. Salah 5.1│                  │
│  │ Team     ▼  │  │ Players  50 │  │ 2. Haaland │                  │
│  │ £4.0-£15.0  │  │ GW       22 │  │ 3. Saka    │                  │
│  └─────────────┘  └─────────────┘  └─────────────┘                  │
│                                                                      │
│  ┌──────────────────────────────────────────────────────────────┐   │
│  │ Player Table                                                 │   │
│  │ Name       | Team | Pos | Price | Form | xPTS | Goal%        │   │
│  │ ────────────────────────────────────────────────────────────│   │
│  │ Salah      | LIV  | MID | 12.8  | 8.2  | 5.1  | 45%         │   │
│  │ Haaland    | MCI  | FWD | 14.2  | 9.1  | 5.0  | 52%         │   │
│  └──────────────────────────────────────────────────────────────┘   │
└──────────────────────────────────────────────────────────────────────┘
```

2. Team Optimizer Page

```
┌──────────────────────────────────────────────────────────────────────┐
│ 🏆 Team Selector - Optimal Squad Builder                             │
├──────────────────────────────────────────────────────────────────────┤
│ Budget: £100.0M  |  Constraints: [2 GKP, 5 DEF, 5 MID, 5 FWD]       │
│                                                                      │
│  ┌──────────────────────────┐  ┌──────────────────────────────┐     │
│  │ Optimal Squad (4-3-3)    │  │ Bench Options                │     │
│  │                          │  │                              │     │
│  │ GKP: Ramsdale (£5.0)     │  │ GKP: Arek (£4.2)             │     │
│  │ DEF: Saliba, Gabriel,    │  │ DEF: Reguilon, Botman        │     │
│  │      Porro, Estupinan    │  │                              │     │
│  │ MID: Salah, Saka,        │  │ Bench: Eze, Mac Allister     │     │
│  │      Palmer, Bruno       │  │                              │     │
│  │ FWD: Haaland, Watkins    │  │ Expected Points: 78.4        │     │
```

```
│  └──────────────────────┘  └──────────────────────────────┘│
│                                                            │
│  [Optimize Squad]  [Lock Team]  [Export]                   │
└────────────────────────────────────────────────────────────┘
```

Data Flow in Streamlit



Streamlit Flow Diagram

Caching Strategy

All Supabase queries use `@st.cache_data` with 1-hour TTL:

```python
@st.cache_data(ttl=3600)
def load_players() -> pl.DataFrame:
    """Load player data - cached for 1 hour."""
    return read_from_supabase("silver_players", ...)
```

Summary

Aspect	Current State	Future State
Data	FPL, Vaastav, Understat	+ News APIs, Social Media
ML	LightGBM regression/ classification	+ Deep learning, Transformer models

Aspect	Current State	Future State
AI	Statistical simulation	+ Sentiment analysis, LLM insights
UI	Streamlit dashboard	+ Mobile app, Slack bot

The project provides a solid foundation for fantasy sports prediction and can be extended with additional AI capabilities for enhanced decision support.